

NAMAN YESHWANTH KUMAR

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SUMMARY

Full-stack engineer (MS Computer Engineering, ASU — GPA 3.81, May 2026) who solo-ships AI-powered platforms to real users. Built Trezzit (app.trezzit.com), a production financial-workflow SaaS serving **600+ users** for **12+ months** — with an MCP server that lets AI assistants take actions across the platform (create bills, query expenses, get balances) and an AI receipt pipeline iterated from **60% → 95%+** accuracy through semantic validation. Built SiliconCrew, a 21-tool LangGraph agent with multi-provider LLM routing and autonomous self-correction.

TECHNICAL SKILLS

Full-Stack & TypeScript: React 19 | TypeScript | Next.js | Node.js | Material-UI | Tailwind CSS | PWA | Python | Django REST Framework | FastAPI | PostgreSQL | Redis | Celery | WebSocket | REST API Design

AI Agents & Tooling: LangGraph | LangChain | MCP (Model Context Protocol) | Tool Use / Function Calling | Multi-provider LLM Routing (Gemini / OpenAI / Claude) | RAG Pipelines | Pydantic Output Validation | Cursor | Claude Code

Cloud & Infra: AWS | Docker | DigitalOcean | Vercel | Supabase | Nginx | GitHub Actions CI/CD | systemd | Linux | Git

PROJECT EXPERIENCE

Trezzit — Production Financial-Workflow SaaS, Sole Engineer (app.trezzit.com)

Feb 2025 – Present

- Solo-built and shipped Trezzit, a production bill-splitting / expense-reconciliation SaaS serving **600+ users** for 12+ months — **341 React components**, **60+ REST endpoints**, **40 DB models**, **995+ backend tests**; Django 5.1 + DRF, React 19 PWA, PostgreSQL (Supabase), Redis, Celery, AWS Lightsail (staging) + DigitalOcean VPS (prod), Vercel; GitHub Actions CI/CD in ~3 min, prod→staging DB sync, encrypted DB backups.
- Built a two-stage AI receipt pipeline (Gemini, OpenAI, Claude, Grok) iterated from ~60% → 95%+ parsing accuracy — semantic validation gate with automatic fallback to higher-capability model on failure; pre-computed balance system with O(1) reads, zero-sum invariant enforcement, optimistic locking on bills, and full audit trail (BalanceUpdateLog + ValidationSnapshot) — auditability and accuracy treated as non-negotiable.
- Built an **MCP server** that lets AI assistants take actions across the platform — create bills, create settlements, query balances and expenses, search bills, upload receipts — over the same auth and data layer as the web app; production observability tracks per-endpoint p95 latency and per-provider LLM cost/tokens across 60+ endpoints.

SiliconCrew — 21-Tool LangGraph Agent with Cross-Tool Actions & Self-Correction

Nov 2025 – Present

- Built a LangGraph ReAct agent with **21 tools** across 6 categories that drives an end-to-end workflow from natural language input to a verified, deployed artifact; self-corrects on failure (read diagnostic → identify root cause → patch → re-verify) with zero human input; achieved **37/92 first-pass on CVDP** and **5/5 on ASU Spec2Tapeout**; multi-provider LLM routing (Gemini / OpenAI / Anthropic), full MCP server (stdio/SSE/HTTP) compatible with Claude Desktop, VS Code, and any MCP client; FastAPI + WebSocket + Next.js with real-time tool-call streaming.

HEAL.AI — AI Billing Assistant with RAG & Workflow Automation

Sep – Nov 2025

- Won Devlabs Hackathon (1st place)** with HEAL.AI, an AI billing assistant; built a RAG pipeline — PDF chunking, embeddings, cosine-similarity retrieval over uploaded insurance policies — enabling natural-language Q&A, billing overcharge detection, and automated dispute-email generation via Gemini 2.5; FastAPI + Next.js + TypeScript, Dockerized.

Intel Automated Self-Checkout — Open Source Contribution

Jan – Feb 2025

- Contributed sensor analytics microservice to **Intel's official self-checkout reference architecture** — configurable data pipeline (MQTT, Kafka, HTTP) for real-time LiDAR and weight sensor data with Grafana dashboard; **PR #655 merged upstream** after Intel engineering review.

WORK EXPERIENCE

Software Developer (Sole Engineer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Engineered a production Django platform (PostgreSQL + Celery + Nginx on DigitalOcean) processing **18,000+ orders over 3 years**; built 3 role-based portals (admin, artisan, customer) with multi-method auth — automated PDF invoice generation and WhatsApp Business API delivery on every state transition, eliminating ~20 min of manual relay per order.
- Reverse-engineered third-party Jewely ERP (**zero public API**) through direct SQL schema analysis; built bidirectional bridge model syncing production weights and ERP status in real time; engineered inbound WhatsApp state-machine chatbot (JSON state per user, admin + customer flows) — full accountability loop across all three portals without disrupting legacy system.

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University — Coursework: Artificial Intelligence · Foundations of Algorithms · Reinforcement Learning in Robotics · Perception in Robotics · Algorithm/Hardware Co-Design

Tempe, AZ

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

AWARDS & HONORS

- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI billing assistant with a custom RAG pipeline (768-dim embeddings, cosine-similarity retrieval) for insurance Q&A and automated dispute-email generation.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.